RESEARCH QUESTION

HYPOTHESIS

OBJECTIVE

Objectives, Questions & Hypotheses

All you need to know!



LEARNING OBJECTIVES

- Understand the importance of a good research question
- Learn how to identify a research gap
- Explore methods to develop a research question
- Apply the PICOT and FINER criteria
- Practice framing sample questions

WHY IS THE RESEARCH QUESTION IMPORTANT?

- It is the **foundation** of any research project
- Guides the **research design, data collection, and analysis**
- Helps in **staying focused and avoiding vague studies**
- Ensures the research is **relevant, feasible, and ethical**



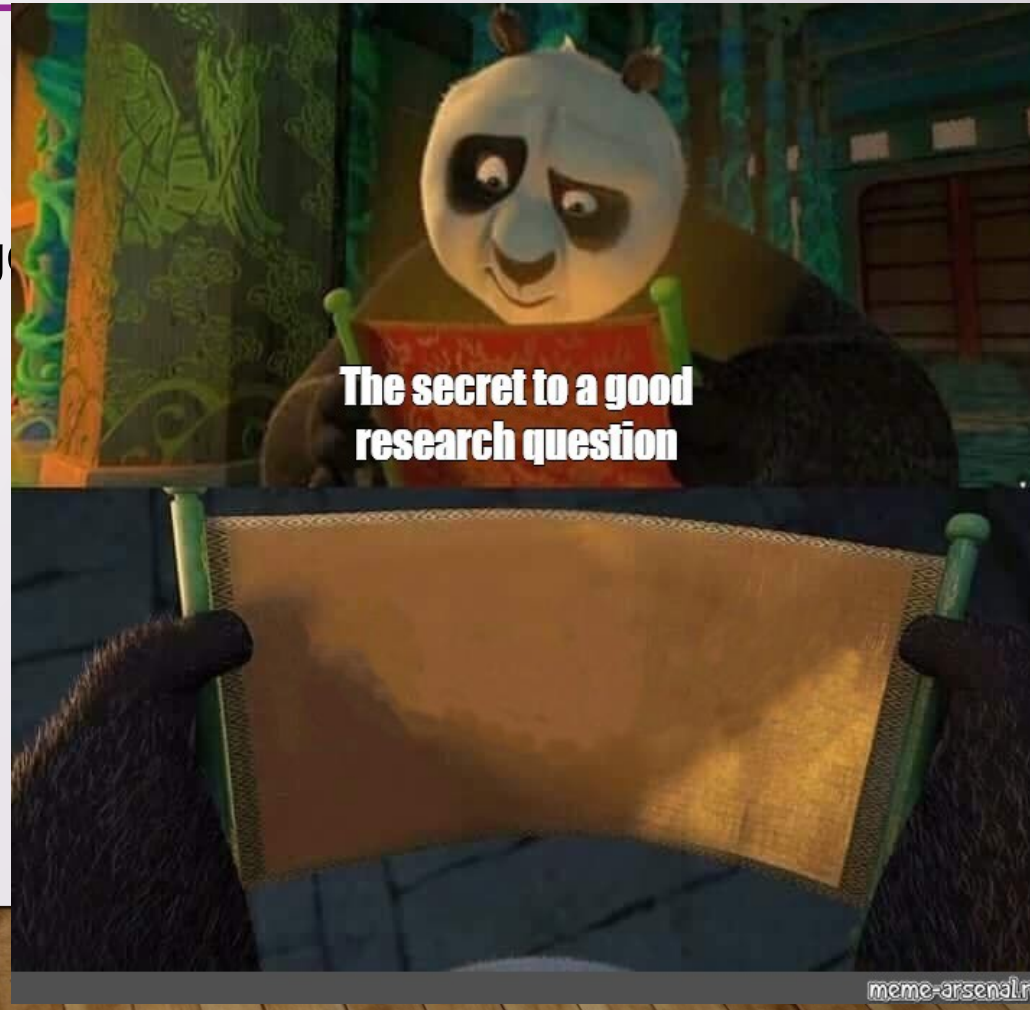
WHAT IS A RESEARCH QUESTION?

- A **clear, focused, concise, and arguable question** around a specific issue
- Drives the research objective
- Must be answerable through **data or investigation**



Characteristics of a good research question (FINER)

- **F – Feasible** (Can you do it?)
- **I – Interesting** (To you and others)
- **N – Novel** (New, or adds to existing knowledg
- **E – Ethical** (Does not harm participants)
- **R – Relevant** (To field and future practice)



HOW TO CHOOSE A RESEARCH TOPIC

- Think of areas **you find interesting in dentistry**
- Look at **common problems or gaps in clinical practice**
- Read current **journals, articles, or case studies**
- Talk to **teachers or seniors**
- Choose something **narrow and manageable**

Me: "I don't wanna read a scientific paper, they're so boring and-"

What We Know So Far: Answering the Question of What, How, and Why: *exists*

Me:



METHODS TO FRAME A RESEARCH QUESTION

- Use frameworks like:
- **PICOT** for clinical questions
  - P = Population
  - I = Intervention
  - C = Comparison
  - O = Outcome
  - T = Time (optional)
- **PICO Example:**
  In teenagers with dental caries (P), is fluoride toothpaste (I) more effective than herbal toothpaste (C) in reducing caries (O)?

TYPES OF RESEARCH QUESTIONS

- **Descriptive:** What is happening?
- **Comparative:** Is there a difference?
- **Analytical/Causal:** Why is it happening?
- **Exploratory:** What are the factors?

- **Common Mistakes to Avoid**
- 🚫 Too broad or too vague
  🚫 Unanswerable questions
  🚫 Choosing a topic without checking feasibility
  🚫 Ignoring existing literature
  🚫 Overly ambitious in scope

- **Examples – Good vs. Poor Questions**
- **Poor:** “Is dental health important?”
  Better: “Does regular scaling reduce gingivitis in college students aged 18–22?”

- **Interactive Activity**

Think-Pair-Share:

Formulate a research question based on:

- **Dental caries in children**
- **Use of mouthwashes**
- **Smoking and oral lesions**

  Discuss with your partner and refine it

Summary

Research question is the **starting point**

Should be **clear, focused, and relevant**

Use **frameworks like FINER & PICOT**

Practice and feedback are key to improvement

- **Final Thoughts**

“The right question can lead to a discovery that changes practice.”

Encourage curiosity, critical thinking, and clinical relevance in every research journey

WHAT IS A HYPOTHESIS?

- A **hypothesis** is an **educated guess** or **proposed explanation** for a phenomenon
- It is **based on prior knowledge, observation, or literature**
- It must be **testable and falsifiable**
- Example:
  Using fluoride toothpaste reduces dental caries in children.

CHARACTERISTICS OF A GOOD HYPOTHESIS

- ✅ Clear and specific
  ✅ Testable with available methods
  ✅ Based on previous knowledge or observations
  ✅ Predicts a relationship between variables



TYPES OF HYPOTHESES

- **Null Hypothesis (H_0):**
  - No difference or relationship exists
  - e.g., *There is no difference in caries incidence between children using fluoride and herbal toothpaste.*
- **Alternative Hypothesis (H_1):**
  - A difference or relationship does exist
  - e.g., *Children using fluoride toothpaste have fewer caries than those using herbal toothpaste.*

WHAT ARE RESEARCH OBJECTIVES?

- Research objectives state **what you aim to achieve**
- They guide the **methodology and data collection**
- Should be **SMART:**
  - **S**pecific
  - **M**easurable
  - **A**chievable
  - **R**ealistic
  - **T**ime-bound

DIFFERENCE – AIM VS. OBJECTIVES

- **Aim:** Broad overall goal

  To assess the effectiveness of fluoride toothpaste in preventing caries

- **Objectives:** Narrow, actionable steps
  - To compare caries prevalence in users of fluoride vs. non-fluoride toothpaste
  - To measure caries reduction after 3 months of use

HOW TO WRITE RESEARCH OBJECTIVES

- Use action verbs:
  - To compare
  - To assess
  - To evaluate
  - To identify
  - To determine
- Example:
- **Objective:** *To evaluate the antimicrobial effect of chlorhexidine mouthwash on oral biofilm.*

LINKING HYPOTHESIS WITH OBJECTIVES

- **Hypothesis:**
  Chlorhexidine mouthwash significantly reduces oral biofilm.
  Objectives:
- To measure biofilm score before and after use
- To compare biofilm reduction with and without mouthwash

- **Common Mistakes to Avoid**
- Vague or too broad objectives
  Unmeasurable outcomes
  Writing hypothesis as a question
  Mixing aim, objectives, and hypothesis together

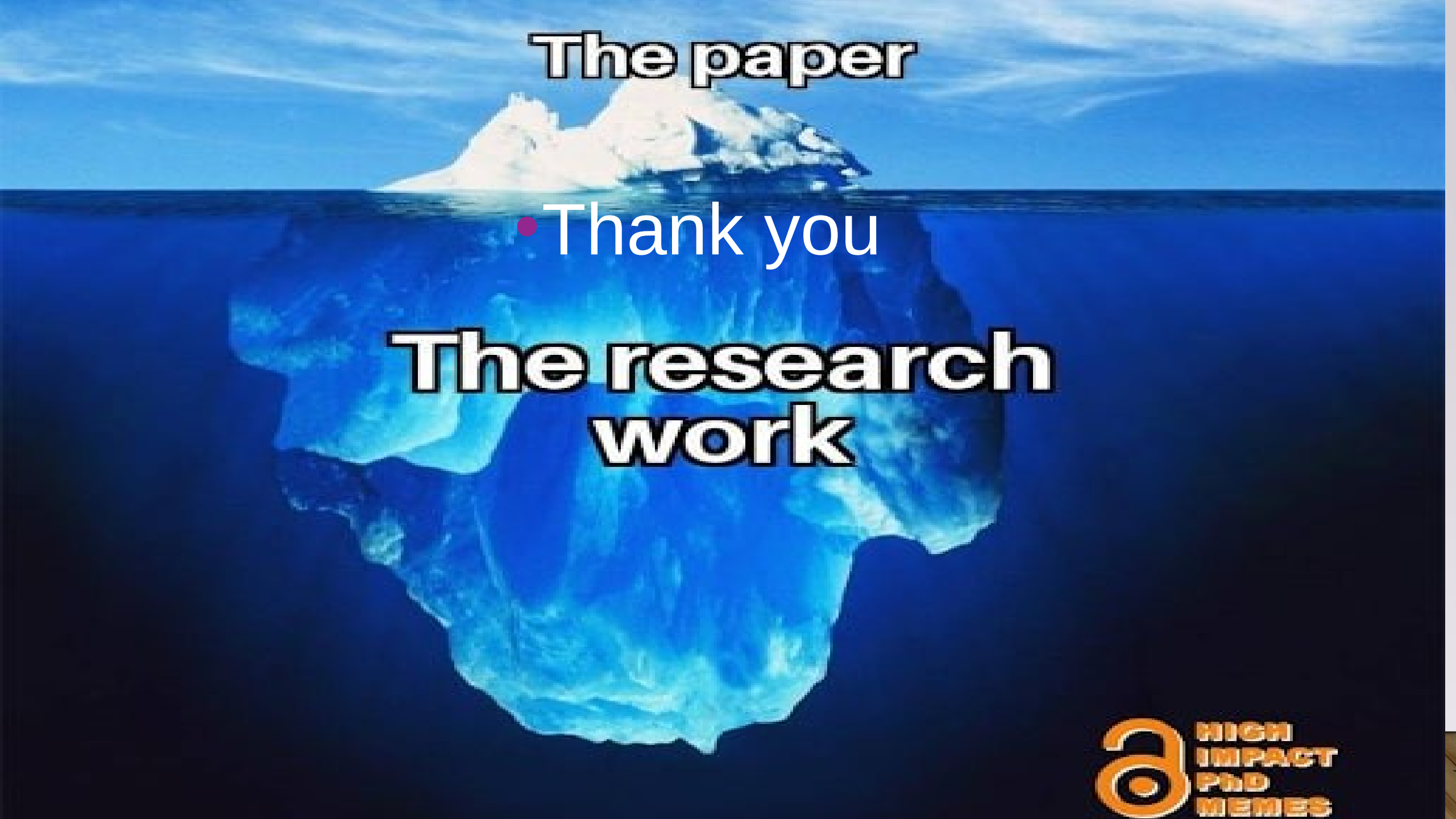
INTERACTIVE ACTIVITY

- **Activity:**
  Choose a dental topic.
- Write 1 hypothesis
- Write 2 related objectives
  Share with a partner and get feedback

- **Summary**
- ✅ Hypothesis = Testable prediction
  ✅ Objectives = Step-by-step research goals
  ✅ Both must be clear, relevant, and researchable
  ✅ Essential for a strong research foundation

- **Final Takeaway**

“Well-formed questions and clear objectives are the roots of every good study.” Keep practicing. Start small. Think logically.



- Thank you